



DEPARTMENT OF MATHEMATICS
ACADEMIC YEAR: 2022-23 EVEN SEMESTER (BMAE201)
QUESTION BANK

MODULE-4: NUMERICAL METHODS -1

1. The area of a circle (A) corresponding to diameter (D) is given below:

D	80	85	90	95	100
A	5026	5674	6362	7088	7854

Find the area corresponding to diameter 105 using interpolation formula.

2. A survey conducted in a slum locality reveals the following information as classified below:

Income per day	Under 10	10-20	20-30	30-40	40-50
No of persons	20	45	115	210	115

Estimate the probable number of persons in the income group of 20 to 25.

3. Given $\sin 45^\circ = 0.7071$, $\sin 50^\circ = 0.7660$, $\sin 55^\circ = 0.8192$, $\sin 60^\circ = 0.8660$,
Find $\sin 57^\circ$ using an appropriate interpolation formula.

4. The population for a town is given by the following table

Year	1951	1961	1971	1981	1991
Population in thousand	19.96	39.65	58.81	77.21	94.61

Using newton's forward and backward interpolation formula, calculate the increase in the population from the year 1955 to 1985.

5. Given $f(40) = 184$, $f(50) = 204$, $f(60) = 226$, $f(70) = 250$, $f(80) = 276$,
 $f(90) = 304$, find $f(38)$ and $f(58)$ using suitable interpolation formulae.
6. From the following table find the number of students who have given
a) Less than 45 marks b) between 40 and 45 marks

Marks	30-40	40-50	50-60	60-70	70-80
No of students	31	42	51	35	31

7. Find $f(9)$ from the data by Newton's divided difference formula:

x	5	7	11	13	17
F(x)	150	392	1452	2366	5202

8. Construct the interpolating polynomial in x for the following data using newton's divided difference formula.

X	2	4	5	6	8	10
y	10	96	196	350	868	1746

9. Determine $f(x)$ as a polynomial in x for the following data using newton's divided difference formula.

X	-4	-1	0	2	5
y	1245	33	5	9	1335

10. Use Lagrange's interpolation formula to find y at $x = 4$, for the given data:

X	0	1	2	5
y	2	3	12	147

11. Use Langrange's interpolation formula to fit a polynomial for the data

X	0	1	3	4
y	-12	0	6	12

Hence estimate y at $x = 2$.



12. Find the real root of $x \log_{10} x - 1.2 = 0$ by Regula- falsi method correct to four decimals.
13. Find the real root of $xe^x = 3$ by Regula- false method correct to four decimals.
14. Find an approximate root of the equation $x^3 - 3x + 4 = 0$ using method of false position, correct to three decimal places which lie between -3 and -2.
15. Find the real root of the equation $x \sin x + \cos x = 0$, near $x = \pi$ using Newton-Raphson method. Carry out 3 iterations.
16. Find the real root of the equation $3x = \cos x + 1$ correct to 3 decimal places using Newton- Raphson method.
17. Evaluate $\int_0^{\pi/2} \sqrt{\sin x} \, dx$ using Simpsons 1/3rd rule, taking 10 equal parts.
18. Evaluate $\int_0^2 \frac{dx}{16+x^2}$ by applying Trapezoidal rule by taking six equal parts.
19. Evaluate $\int_0^5 \frac{1}{4x+5} \, dx$ by using Trapezoidal rule by taking 11 ordinates.
20. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ Simpson's 3/8th rule by taking 6 sub-intervals.
21. Using Simpson's 3/8th rule evaluate $\int_0^1 \frac{1}{1+x} \, dx$ by taking 7 ordinates.
22. Evaluate $\int_0^{\pi/2} \cos x \, dx$ by applying Trapezoidal rule, taking eleven ordinates.
23. Use Simpson's 1/3rd rule with seven ordinates to evaluate $\int_2^8 \frac{dx}{\log_{10} x}$.
24. Use Simpson's 3/8th rule taking 3 equal parts to evaluate $\int_0^{0.3} (1-8x^3)^{\frac{1}{2}} \, dx$.
25. Evaluate $\int_4^{5.2} \log_e x \, dx$ using Simpson's (1/3)rd rule, taking 6 equal parts.
26. Use Simpson's 3/8rd rule to find $\int_0^{0.6} e^{-x^2} \, dx$ by taking 6 sub-intervals.